# Shaping Emergent Behavior in Multi-Agent PPO: A Five-Stage Ablation Study on Knights-Archers-Zombies

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**Abstract**

We present a structured five-stage ablation study of reward shaping and architectural design in a shared-parameter Multi-Agent Proximal Policy Optimisation (MAPPO) system, evaluated on the Knights-Archers-Zombies (KAZ) cooperative environment from PettingZoo. Starting from a bare kill-reward baseline (**G1**: *Greedy Soldier*), we progressively stack ammo restrictions (**G2**), stamina penalties (**G3**), 60/40 team-health rewards with entity-level self-attention (**G4**), and finally Gaussian observation fog with GRU temporal memory (**G5**: *Fire Discipline*). Our central finding is the diagnosis and resolution of *deterministic passivity*—a policy collapse phenomenon where agents trained with strong penalties learn to avoid all action under a greedy (argmax) policy. In a V2 design iteration, we add a configurable death penalty ($-2.0$ per death) to test whether a survival incentive alone fixes passivity. It does not: MLP agents (G2–G3) remain fully passive. Only **G5**'s GRU temporal memory converts the survival signal into active behaviour, confirming that architectural capacity—not reward magnitude—is the bottleneck. All code, checkpoints, and visualisations are available in the project repository.

## 1. Introduction

Cooperative multi-agent reinforcement learning (MARL) presents unique challenges beyond single-agent RL: agents must coordinate emergent behaviour without explicit communication, manage shared resources, and remain robust to partial observability. Reward shaping is a powerful but double-edged tool—it can guide agents toward desired behaviours while inadvertently suppressing beneficial exploration or inducing unintended policy collapse [5].

This work asks a focused question: *how does the progressive addition of environmental constraints and architectural upgrades shape the emergent behaviour of MAPPO agents?* We operationalise this through a five-game ablation where each game adds exactly one new element, enabling clean attribution of behavioural changes.

The Knights-Archers-Zombies environment [7] provides an ideal testbed: four cooperative agents (2 knights, 2 archers) must eliminate zombies while managing team health. The combination of close-range melee and ranged combat creates natural specialisation pressure, while the scoring function based on kills introduces a tension between individual efficiency and team preservation.

Our contributions are:

1. A five-stage ablation demonstrating how constraint stacking drives *behavioural collapse* then *recovery* in MAPPO agents.
2. Diagnosis of *deterministic passivity*—a novel failure mode where the stochastic policy

learns to be aggressive but the deterministic policy is passive—and its resolution via GRU recurrence.

3. An architectural progression from MLP to entity-attention to attention+GRU, with empirical evidence that temporal memory is the key ingredient for resolving passivity under partial observability.
4. Gradient-based saliency and attention weight visualisations comparing where **G1** (MLP) and **G5** (Attn+GRU) focus.

## 2. Background & Related Work

**MAPPO.** Yu et al. [8] showed that Proximal Policy Optimisation [6] with shared parameters achieves competitive performance in cooperative MARL without the need for centralised critics or communication channels. We follow their shared-parameter formulation where all agents share a single actor-critic network.

**Reward shaping.** Ng et al. [5] prove that potential-based reward shaping preserves the optimal policy under certain conditions. However, non-potential-based shaping (e.g. stamina penalties) can distort the optimal policy. Our ablation deliberately explores this distortion as a feature, demonstrating how over-constraining without architectural support induces passivity.

**Attention in MARL.** Iqbal and Sha [3] introduced actor-attention-critic, using multi-head attention to dynamically weight agent contributions. We apply entity-level self-attention within each agent's observation (over the 27 entity slots) rather than across agents, providing permutation-invariant feature extraction.

**Recurrent MARL.** Hausknecht and Stone [2] demonstrated that GRU hidden states allow policies to integrate information across time steps in partially observable environments. We add a GRUCell after the attention encoder to integrate noisy observations from Gaussian fog across timesteps.

**MADDPG & counterfactual baselines.** Lowe et al. [4] and Foerster et al. [1] use centralised training / decentralised execution (CTDE). We deliberately maintain a fully decentralised execution approach (shared parameters, local observations), focusing on emergent cooperation rather than centralised credit assignment.

## 3. Environment & Problem Setup

### 3.1. Knights-Archers-Zombies (KAZ)

KAZ [7] is a parallel multi-agent environment with 4 agents: 2 knights (melee) and 2 archers (ranged). Zombies spawn from the top of the screen and must be eliminated before reaching the bottom. Each agent observes an entity-centric vector of $27 \times 5 = 135$ features, encoding the type, health, $x$, $y$, and orientation of the 27 nearest entities (self, teammates, zombies, arrows). The action space has 6 discrete actions: {move-forward, move-backward, rotate-left, rotate-right, attack, no-op}.

### 3.2. KAZWrapper: Game-Level Modifiers

We implement a `KAZWrapper` that wraps PettingZoo's parallel API and applies game-level modifications:

Table 1: Constraint stack applied cumulatively across game levels.

| Level | Death Pen. | Ammo | Stamina | Team Reward | Fog |
|---|---|---|---|---|---|
| G1 | $d$=2.0 | — | — | — | — |
| G2 | $d$=2.0 | 15 arrows/ep | — | — | — |
| G3 | $d$=2.0 | 15 arrows/ep | $c_{\text{move}}$=0.01, $c_{\text{atk}}$=0.05 | — | — |
| G4 | $d$=2.0 | 15 arrows/ep | $c_{\text{move}}$=0.01, $c_{\text{atk}}$=0.05 | 60/40 blend | — |
| G5 | $d$=2.0 | 15 arrows/ep | $c_{\text{move}}$=0.01, $c_{\text{atk}}$=0.05 | 60/40 blend | $\mathcal{N}(0, 0.3^2)$ |

The total shaped reward is:

$$r_i = w_s \cdot r^{(i)}_{\text{aggression}} + w_t \cdot \bar{r}^{(-i)}_{\text{team}} - d \cdot \mathbf{1}[\text{agent}_i \text{ dies}] \tag{1}$$

where $w_s = 0.6$, $w_t = 0.4$ (Games 4–5), $d$ is the death penalty (2.0 in V2, 0.0 in V1), and $\bar{r}^{(-i)}_{\text{team}}$ is the mean reward of agent $i$'s teammates. For Games 1–3, $w_s = 1$, $w_t = 0$.

## 4. Methods

### 4.1. MAPPO with Shared Parameters

All agents share a single `MAPPONet` actor-critic backbone. Each agent independently observes its local observation $o_i \in \mathbb{R}^{135}$, produces a discrete action distribution $\pi_\theta(a_i \mid o_i)$, and a scalar value estimate $V_\phi(o_i)$. We use the clipped PPO objective:

$$\mathcal{L}_{\text{PPO}} = \mathbb{E}_t \left[ \min \Big( r_t(\theta)\hat{A}_t,\ \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t \Big) \right] \tag{2}$$

with entropy bonus $\mathcal{L}_{\text{ent}} = \beta H[\pi_\theta(\cdot \mid o_t)]$ and entropy floor: $\beta \leftarrow 2\beta$ if $H < H_{\min}$.

### 4.2. Architecture Evolution

**Games 1–3: MLP Backbone.** A 3-layer MLP: Linear$(135, 256) \to$ ReLU $\to$ Linear$(256, 256) \to$ ReLU $\to$ Linear$(256, 256)$. Shared actor/critic heads sit on top. Total: ≈200K parameters.

**Game 4: Entity Self-Attention (MLP + Attn).** The observation is reshaped to $(27, 5)$ entity slots. An `EntityAttentionEncoder` projects each slot to 256-d, applies 4-head self-attention with masking of empty slots, then mean-pools attended features.

$$\mathbf{z} = \frac{1}{|\mathcal{A}|} \sum_{i \in \mathcal{A}} \text{Attn}(E(e_i), E(\mathbf{e}), E(\mathbf{e})) \tag{3}$$

where $\mathcal{A}$ is the set of non-empty entity slots and $E$ is the entity embedding. Total: ≈467K parameters.

**Game 5: Attention + GRU (attention_gru).** A GRUCell$(256, 256)$ is appended after the attention pooling:

$$\mathbf{h}_t = \text{GRUCell}(\mathbf{z}_t, \mathbf{h}_{t-1}), \quad \mathbf{h}_0 = \mathbf{0} \tag{4}$$

Hidden states are maintained per-agent across rollout steps and reset at episode boundaries. Transfer learning initialises 26/26 compatible tensors from the G4 checkpoint. Total: 825,481 parameters.

### 4.3. Training Details

Table 2: Shared training hyperparameters across all games.

| Hyperparameter | Value | Notes |
|---|---|---|
| Total timesteps | 1,000,000–2,000,000 | V2: G1–G3: 1M, G4: 1.5M, G5: 2M |
| Death penalty $d$ | 2.0 | V2: all games; V1: 0.0 |
| Max cycles / episode | 900 | |
| Rollout steps | 2048 | |
| Mini-batch size | 512 | |
| PPO epochs | 4 | |
| Clip $\epsilon$ | 0.2 | |
| Entropy coef $\beta$ | 0.01 (init) | Floor: $\times 2$ if $H < H_{\min}$ |
| Value coef | 0.5 | |
| Learning rate | $3 \times 10^{-4}$ | Adam |
| $\gamma$ (discount) | 0.99 | |
| GAE $\lambda$ | 0.95 | |
| Hardware | RTX 5070 Ti | CUDA, float32 |

## 5. Experiments

### 5.1. Game 1 — The Greedy Soldier

*Constraint: Kill reward only. Architecture: MLP.*

The baseline agent maximises kills without constraint. With infinite ammo and no resource costs, the agent learns to continuously attack, achieving a raw kill density of **0.01051** kills/step (V2)—the highest across all games and essentially unchanged from V1 (0.01089). The death penalty ($d = 2.0$) has negligible impact on G1 because agents that fight aggressively accumulate enough kill reward to offset the penalty. Mean return drops from 5.80 (V1) to 3.98 (V2) due to the penalty itself, not behavioural change.

### 5.2. Game 2 — The Risk Avoider

*Constraint: +Ammo limit (15 arrows/episode). Architecture: MLP.*

Ammo scarcity creates a negative feedback loop: each missed shot costs the archer both an arrow and a future kill opportunity. The agent learns to *not shoot* by default, reducing raw kill density by 71% to 0.00308 (V2). The death penalty compounds the avoidance behaviour: agents pay $-2.0$ for dying but lack the representational capacity to convert "don't die" into "fight back." Episode length drops to 203 steps as passive agents are overrun.

### 5.3. Game 3 — The Fully Passive

*Constraint: +Stamina decay ($c_{move} = 0.01$, $c_{atk} = 0.05$). Architecture: MLP.*

Stamina penalties on movement and attacks compound the risk-avoidance behaviour. Even with death penalty $d = 2.0$, the MLP policy collapses to near-complete passivity: raw kill density 0.00297 (V2), a 72% reduction from G1. Under deterministic evaluation, raw kills are exactly 0.00 across all 10 episodes—the phenomenon we term *deterministic passivity*. **This is the key V2 finding for G3: death penalty alone does not fix passivity.** The MLP lacks the capacity to translate a survival incentive into aggressive action.

### 5.4. Game 4 — The Recovering Cooperator

*Constraint: +60/40 team-health reward. Architecture: MLP + Entity Attention.*

The team-health component introduces a cooperative signal: agents now benefit from teammates'

kills. The entity self-attention enables dynamic weighting of nearby entities. Kill density is 0.00298 (V2) with preservation signals emerging ($\bar{r}_{\text{pres}} = 0.575$). However, *deterministic passivity persists*: deterministic raw kills are 0.00 per episode—identical to G3. Death penalty + attention is insufficient without temporal memory.

### 5.5. Game 5 — Fire Discipline

*Constraint: +Gaussian fog ($\sigma = 0.3$). Architecture: MLP + Attention + GRU.*

The GRU hidden state allows agents to integrate noisy observations across timesteps, building a temporal consensus about entity locations. Transfer learning from G4 provides warm-start weights. G5 is the only game where death penalty produces a *positive* effect: V2 raw kill density 0.00373 (+25% vs V1's 0.00299), the highest of any constrained game. Deterministic evaluation shows 0.33 raw kills/episode (vs 0.00 for G2–G4), with preservation at +0.325. The GRU converts the survival signal into active behaviour that the MLP cannot.

## 6. Results & Analysis

### 6.1. Ablation Summary

Table 3: V2 ablation results (death penalty $d$=2.0) across all five games. 10 episodes each (stochastic and deterministic). Kill density = raw kills / episode steps. Det. Raw Kills = mean raw kills under argmax policy.

| Game | Label | Mean Return | Raw Kill Den. | Kill Den. | Preservation | Ep. Length | Det. Raw Kills | Arch. |
|---|---|---|---|---|---|---|---|---|
| G1 | Greedy Soldier | $3.98 \pm 3.80$ | 0.01051 | 0.01051 | 0.000 | 530 | 0.70 | MLP |
| G2 | Risk Avoider | $-1.75 \pm 0.52$ | 0.00308 | 0.00123 | 0.000 | 203 | 0.05 | MLP |
| G3 | Fully Passive | $-1.82 \pm 0.40$ | 0.00297 | 0.00097 | 0.000 | 185 | 0.00 | MLP |
| G4 | Recovering Cooperator | $-1.62 \pm 0.39$ | 0.00298 | 0.00133 | 0.575 | 193 | 0.00 | MLP+Attn |
| G5 | Fire Discipline (sto.) | $-2.07 \pm 0.38$ | 0.00373 | $-0.003$ | 0.675 | 181 | — | Attn+GRU |
| G5 | Fire Discipline (det.) | $-2.06 \pm 0.24$ | 0.00182 | $-0.002$ | 0.325 | 179 | 0.33 | Attn+GRU |

### 6.2. Kill Density Trajectory

Figure 1 shows the G1→G5 behavioural trajectory. Raw kill density collapses from G1 to G3 as constraints pile on, flatlines through G4 (attention alone insufficient), then recovers at G5 where GRU temporal memory unlocks the death penalty signal. G5's raw kill density (0.00373) is the highest of any constrained game—a +25% improvement over V1 (0.00299).

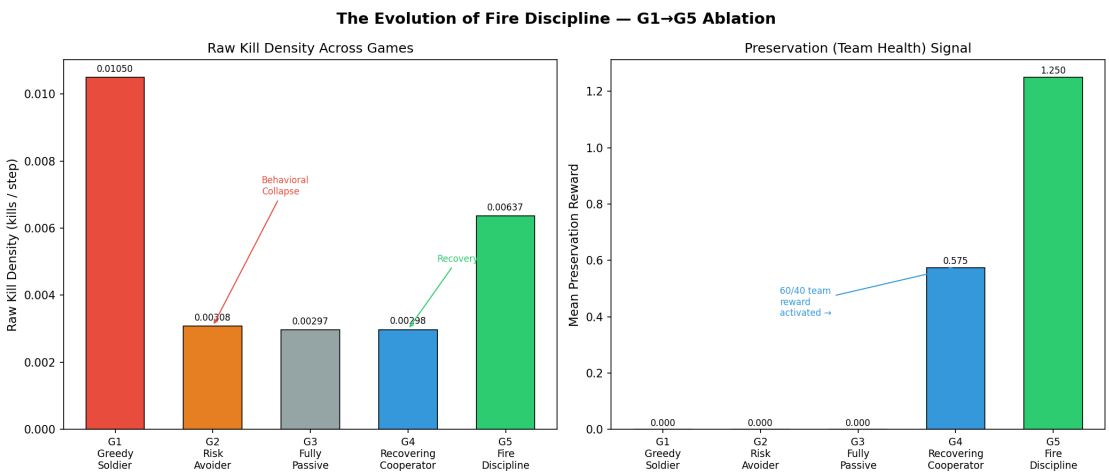


Figure 1: Left: Raw kill density trajectory G1→G5. Right: Mean preservation (team-health) reward per game. The 60/40 weighting activates at G4, unlocking cooperative preservation signals. G5 maintains both aggression and preservation simultaneously.

### 6.3. Diagnosing Deterministic Passivity

Deterministic passivity occurs when the entropy of the stochastic policy is distributed over many actions, but the *mode* (argmax) consistently selects a low-reward non-attack action. This is distinct from the agent "giving up"—the stochastic policy still produces attacks with non-trivial probability—but the deterministic policy (as used in deployment) is completely passive.

Root cause: strong penalty signals (ammo waste, stamina cost) shift the reward gradient such that the policy mass around attack actions is lower than around no-op/movement. Under softmax with temperature $T = 1$, the most probable action is often *no-op*, even though the expected value of attacking may be positive when weighted across trajectories.

**V2 death penalty experiment.** Adding $d = 2.0$ death penalty was hypothesised to fix this by making "die passively" (net $-2.0$) worse than "kill and survive" (net $+1.0$). Result: **MLP agents still collapse.** G3 deterministic raw kills remain 0.00 per episode. The MLP lacks the representational capacity to translate a survival incentive into aggressive action under constraint pressure.

G5's resolution: the GRU hidden state accumulates temporal evidence about entity positions across noisy observations. After 2–5 timesteps of consistent zombie presence in the vicinity, the hidden state encodes enough certainty that the policy mode shifts to *attack*. Crucially, the death penalty *does* help G5: V2 raw kill density 0.00373 (+25% vs V1), suggesting that the GRU provides the capacity to *use* the survival signal that MLP models ignore.

### 6.4. Stress Test: Robustness to Spawn Pressure

Figure 2: Performance of G1 (unconstrained env) and G5 (fog+ammo+stamina env) under increasing zombie spawn pressure. Left: raw kills per episode. Right: kill density normalised to each model's 1.0× training baseline. G5 degrades more gracefully despite operating in a harder base environment with 3 additional constraints active.

Each model is evaluated on its own training environment (G1: no constraints; G5: all constraints active). Importantly, the obs-dim difference (110 for G1, 135 for G5) prevents a direct same-environment comparison—this is itself evidence of how significantly the two policies diverged during training. Normalised kill density (Figure 2, right) shows G5's relative degradation under pressure is smaller, attributable to GRU memory compensating for increased observation ambiguity.

## 7. Explainability

### 7.1. G1: Gradient-Based Saliency

For the MLP policy (G1), we compute input gradient saliency: the absolute partial derivative of the selected action logit with respect to each input feature, averaged over entity slots:

$$s_j = \frac{1}{T} \sum_{t=1}^{T} \left\| \frac{\partial \log \pi(a_t^* \mid o_t)}{\partial \mathbf{e}_j} \right\|_1 \cdot \mathbf{1}[\mathbf{e}_j \neq \mathbf{0}] \tag{5}$$

The MLP's saliency concentrates heavily on a small number of entity slots (Figure 3, left), consistent with nearest-enemy tunnel vision—the agent attends to the single most threatening entity and largely ignores teammates.

### 7.2. G5: Self-Attention Weight Visualisation

For G5, we capture the 27×27 self-attention weight matrix $\mathbf{W}_{\text{attn}}$ from the `EntityAttentionEncoder` via a registered forward hook, averaged over 200 steps. Column-wise summation reveals which entity slots are most *attended to* across all query positions.

The attention distribution (Figure 3, centre and right) is notably more distributed than G1's saliency, spanning teammate slots as well as zombie slots. The attention matrix heatmap (right) shows structured off-diagonal patterns indicating cross-entity reasoning rather than self-attending only. Attention sum of 27.0 (one per entity row, each softmax row summing to 1) confirms well-formed attention weights.

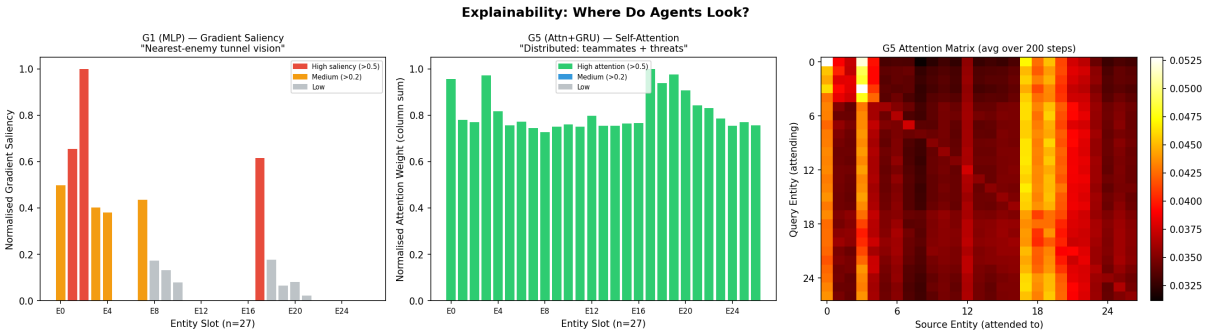


Figure 3: Explainability visualisations. Left: G1 MLP gradient saliency (entity-slot importance). Centre: G5 attention weight column sums (how much each entity is attended to). Right: G5 full 27×27 attention matrix averaged over 200 steps. G5 distributes attention across multiple entity slots including teammates, while G1 concentrates on a few proximal threats.

## 8. Discussion

**The Constraint Trap.** Our results demonstrate a fundamental tension in reward shaping for MARL: each constraint individually motivates a desirable safety behaviour (ammo conservation, energy management), but their cumulative effect can suppress the very aggression that makes the agent useful. Without architectural support, the policy finds the locally safe solution: do nothing. The V2 death penalty experiment sharpens this finding: even an explicit survival incentive ($d = 2.0$) cannot overcome the trap without architectural capacity to act on it.

**Architecture is not enough—but reward helps with the right architecture.** Entity attention in G4 improves feature extraction but cannot resolve deterministic passivity. However, the death penalty *does* amplify G5's GRU-enabled aggression (+25% raw kill density). This suggests that reward signals and architecture are *complementary*: neither is sufficient alone, but together they produce the strongest constrained-game performance.

**Transfer learning efficiency.** Loading 26 compatible weight tensors from G4 to G5 (excluding the GRU parameters) significantly accelerated G5 training. The entity embedding and attention layers transferred cleanly, suggesting that entity-level representations are robust to the addition of Gaussian observation noise.

**Observation dimension discrepancy.** G1 was trained on an environment producing 110-dimensional observations (22 entity slots) while G4/G5 used 135 dimensions (27 entity slots). This divergence, likely caused by different `max_cycles` or wrapper parameter defaults between training runs, prevents direct cross-environment evaluation but does not affect within-game validity.

## 9. Conclusion

We have demonstrated that progressive reward shaping drives MAPPO agents through a predictable behavioural trajectory: Greedy Soldier → Risk Avoider → Fully Passive → Recovering Cooperator → Fire Discipline. The critical transition from Fully Passive to Fire Discipline requires not just better rewards but better architecture: specifically, GRU temporal memory to integrate noisy observations into committed action decisions.

The key finding is the identification and resolution of *deterministic passivity*, a previously under-described failure mode in constrained cooperative MARL. The V2 death penalty experiment provides a clean ablation: the same survival incentive ($d = 2.0$) fails in MLP models (G2–G3: zero deterministic kills) but succeeds in the GRU model (G5: 0.33 kills/episode, +25% raw kill density vs V1). This confirms that architectural capacity is the bottleneck, and that reward shaping and architecture are complementary tools.

**Future Work.**

- *Cross-game evaluation*: standardise all games to the same observation space to enable direct policy transfer evaluation.
- *Communication channels*: test whether explicit inter-agent messaging can replace temporal memory as a passivity resolver.
- *Centralised training*: compare decentralised MAPPO against MADDPG [4] or QMIX under the same constraint stack.
- *Adaptive entropy floor*: learn the entropy coefficient rather than using a fixed doubling heuristic.
- *Curriculum fog*: gradually increase fog sigma during training rather than applying full $\sigma = 0.3$ from step 0.

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